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# CPU Scheduling Visualizer

## *with Prediction Mode*

*Technical Report*

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**Github page link:**

<https://github.com/Marzana-tabassum/CPU-Scheduling-Visualizer>

**Github Folder:**

<https://drive.google.com/file/d/1Dib4-ZLleUMyA2ovkN6BoF9uYH6gClge/view?usp=sharing>

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## Abstract

This report documents the development of the CPU Scheduling Visualizer with Prediction Mode — a feature-rich, interactive desktop application built using Python and Tkinter. The project goes well beyond the scope of a typical OS lab simulator. It implements five scheduling algorithms including a custom-designed hybrid called SMART-SJF, an ML-based prediction engine using scikit-learn, a real-time animated step-by-step simulation, comparative analytics across all algorithms, and an integrated OS chatbot — all wrapped in a professional dark-themed GUI.

The report captures three major technical challenges encountered during the project: designing and validating a novel hybrid scheduling algorithm, integrating machine learning predictions with meaningful criteria, and engineering a frame-accurate animated simulation engine. Each challenge required substantial research, debugging, and creative problem-solving.

## Introduction

CPU scheduling is one of the most fundamental topics in operating systems education. Standard tools visualize FCFS or Round Robin on a static Gantt chart and call it a day. There is rarely any attempt to help students understand when to use which algorithm, how different workloads interact with different scheduling policies, or what the best choice might be for a given process set.

This project was conceived to solve exactly that problem. The CPU Scheduling Visualizer with Prediction Mode is not just a visualizer — it is a learning and decision-support environment. A student or system designer can input a real or hypothetical process set, run any of the five supported scheduling algorithms, compare their performance metrics side by side, watch a step-by-step animated execution, and then ask the built-in ML model which algorithm would statistically perform best for their input. The OS Chatbot adds another layer, letting users ask natural-language questions about scheduling concepts.

What makes this project genuinely unique in its class is the SMART-SJF algorithm — a hybrid scheduling policy designed and implemented from scratch — combined with a prediction mode that uses actual scheduling performance data to train a classifier. No pre-built scheduling library was used; the ML model was trained on simulation outputs rather than on a static dataset.

## System Design and Architecture

The application is structured as a single-file Python program with clearly separated logical modules. The diagrams below show the high-level system architecture, communication flow between components, and how every feature maps to a specific module.

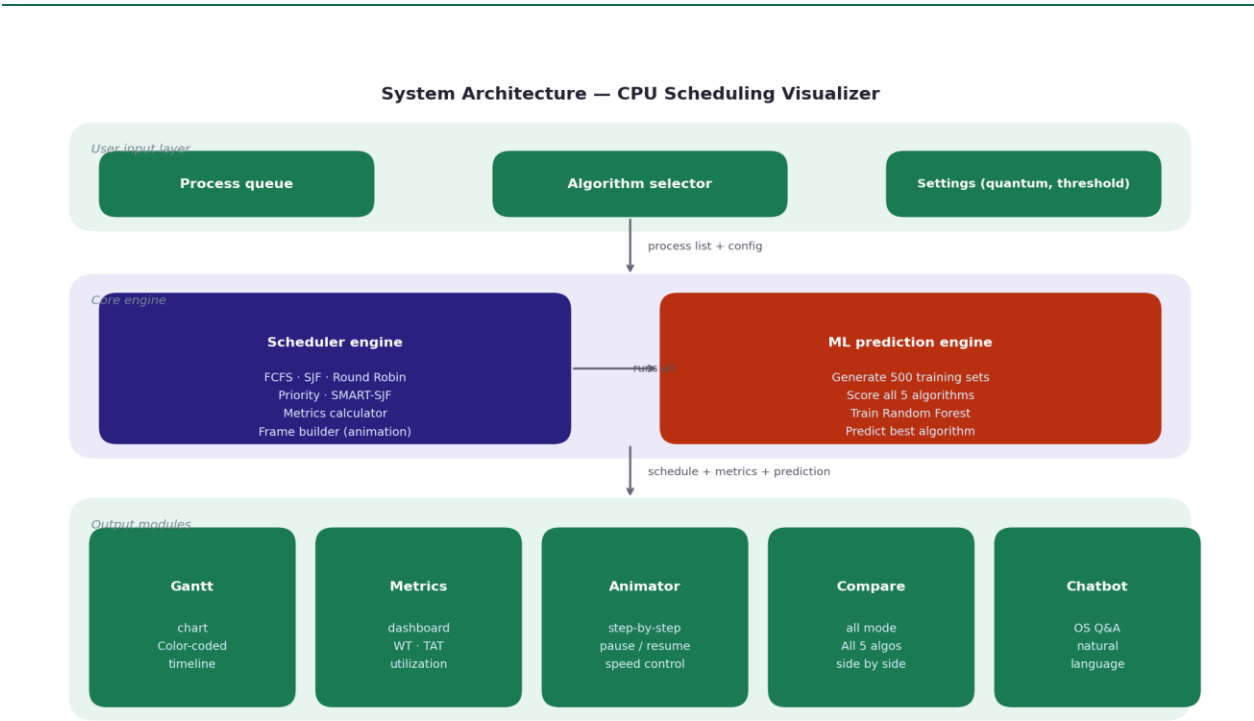


Figure 1: System Architecture — Three-layer overview: User Input → Core Engine (Scheduler + ML) → Output Modules

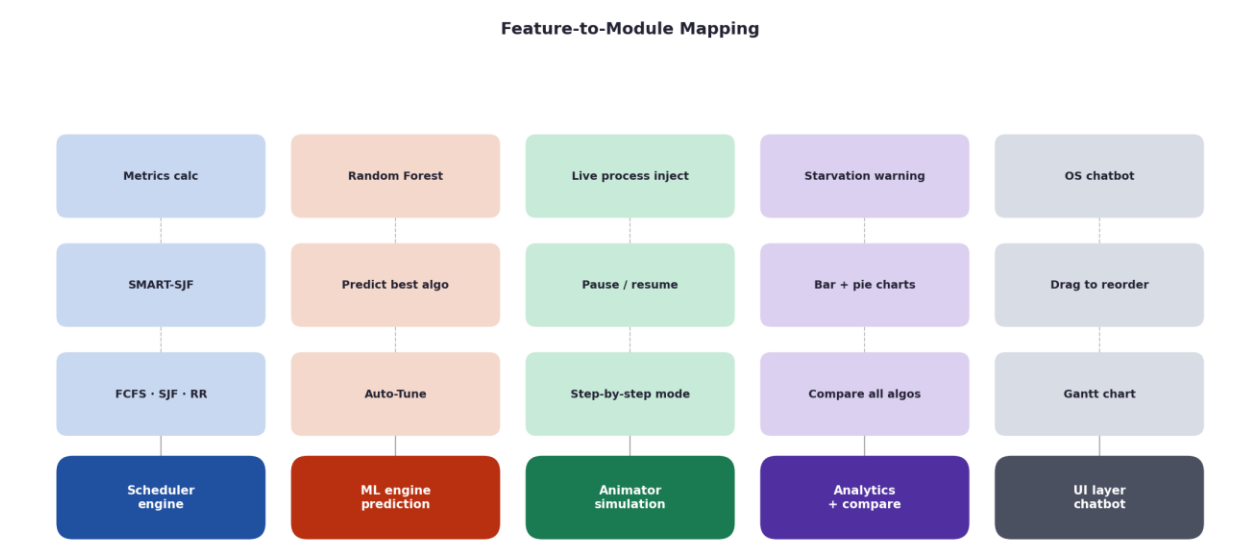


Figure 2: Feature-to-Module Mapping — How every feature maps to a specific module

Architectural Overview

| Layer / Module  | Description                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
| Process Manager | Handles process input, editing, drag-to-reorder priority queue, and live injection of processes mid-simulation. |

| Layer / Module              | Description                                                                                                                        |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scheduler Engine</b>     | Contains pure implementations of FCFS, SJF (non-preemptive), Round Robin, Preemptive Priority, and the custom SMART-SJF algorithm. |
| <b>Gantt Renderer</b>       | Renders color-coded Gantt charts on a Tkinter Canvas with zoom, scroll, and process legend.                                        |
| <b>Metrics Calculator</b>   | Computes per-process and aggregate Waiting Time, Turnaround Time, Response Time, CPU Utilization, and Throughput.                  |
| <b>Animated Simulator</b>   | Frame-by-frame step animation engine with pause/resume/restart controls and live CPU state display.                                |
| <b>ML Prediction Engine</b> | Trains a Random Forest Classifier on simulation outputs and predicts the optimal algorithm for a given process set.                |
| <b>Analytics Dashboard</b>  | Renders bar/pie charts (matplotlib embedded in Tkinter) for visual comparison of algorithm metrics.                                |
| <b>Compare All Module</b>   | Runs all five algorithms on the same input and presents a side-by-side metric table.                                               |
| <b>OS Chatbot</b>           | Keyword-based Q&A engine responding to scheduling, algorithm, and OS concept queries.                                              |

## Key Features

- Five scheduling algorithms including the custom SMART-SJF hybrid, all hand-implemented without external OS libraries.
- Real-time Gantt chart rendering with color-coded processes, scroll/zoom, and a draggable process priority queue.
- ML-based algorithm prediction using Random Forest trained on live simulation outputs — not a static dataset.
- Step-by-step animated simulation with frame-by-frame CPU state visualization, pause, resume, and restart controls.
- Compare All mode that executes all five algorithms on the same process set and produces a unified metrics table.
- Embedded matplotlib analytics with bar and pie chart breakdowns of waiting time, turnaround time, and CPU utilization.
- Live process injection — processes can be added to a running simulation to observe dynamic scheduling behavior.
- Auto-Tune feature that automatically suggests the optimal quantum size for Round Robin based on burst time distribution.
- Starvation detection with configurable threshold and real-time warning when low-priority processes are being starved.
- OS Chatbot that answers natural language questions about scheduling algorithms and OS concepts.

## Challenges Faced

The following section documents three significant engineering challenges encountered during the development of this project..

## Challenge 1: Designing and Validating SMART-SJF — A Novel Hybrid Scheduling Algorithm

### Why SMART-SJF Matters

Standard SJF is provably optimal in terms of average waiting time, but it has a fatal practical flaw: it starves long processes. Round Robin is fair but wastes CPU cycles on context switches and does not account for burst length at all. SMART-SJF was designed to bridge this gap by combining shortest-job preference with aging — a mechanism that gradually elevates the priority of waiting processes over time, ensuring that no process waits forever regardless of its burst length.

The algorithm is particularly well-suited to mixed workloads, where a system has both short interactive tasks and long batch jobs that need guaranteed progress. In environments where fairness and efficiency must coexist — such as interactive operating systems or cloud job schedulers — SMART-SJF offers a meaningful middle ground that neither pure SJF nor Round Robin can provide.

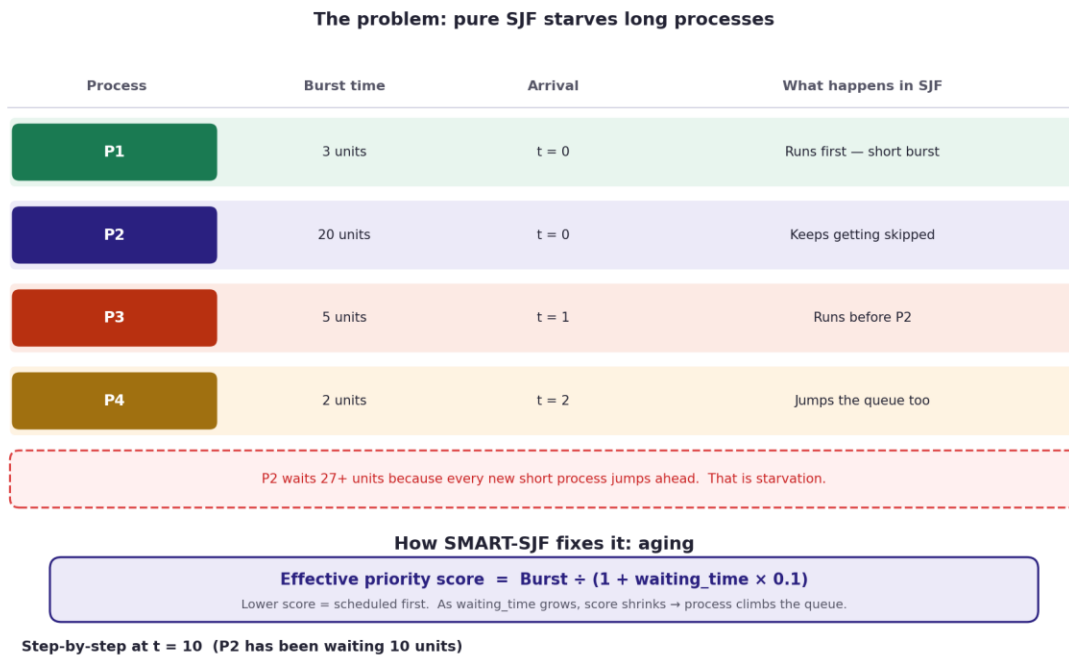


Figure 3: The starvation problem in pure SJF, and how SMART-SJF aging fixes it

**SMART-SJF: Aging Mechanism — Process Scores at t=10 and Over Time**  
**Step-by-step at t = 10 (P2 has been waiting 10 units)**

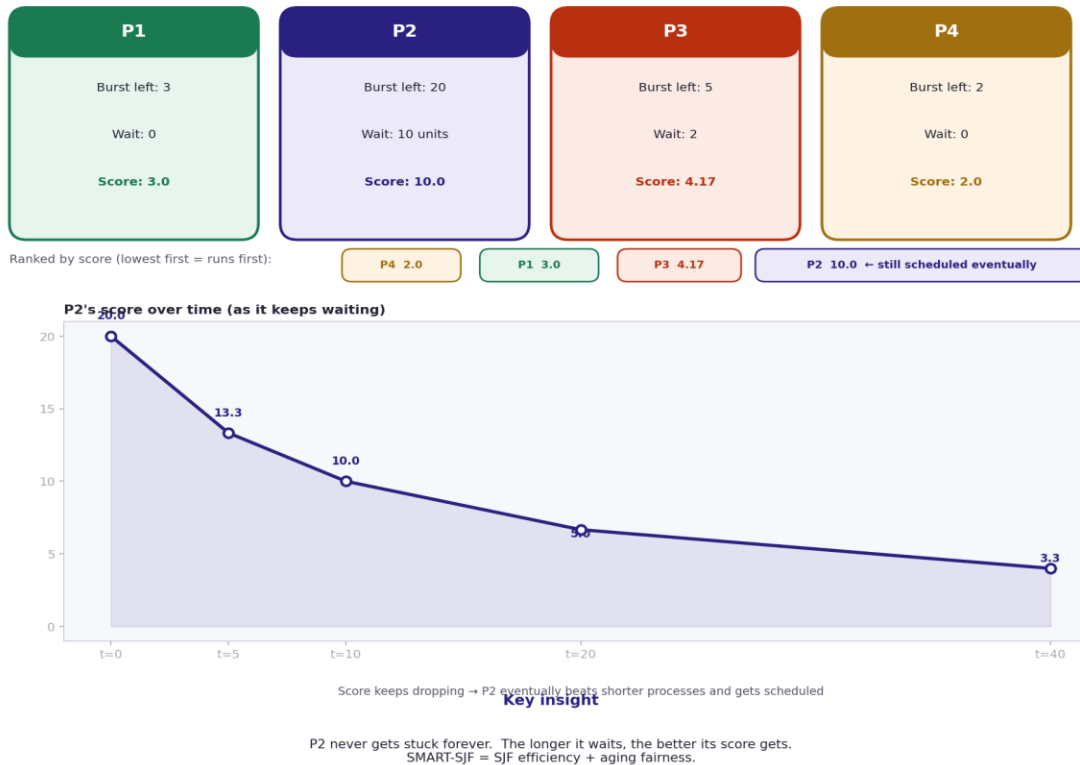


Figure 4: SMART-SJF aging — process scores at t=10 and P2's score declining over time until it gets scheduled

S

**Situation :**

Standard SJF always picks the process with the shortest burst. This works great for average waiting time but creates a real problem: a long process can sit in the queue forever if shorter jobs keep arriving. There was no built-in protection against this. The goal was to build a scheduling algorithm that kept SJF's efficiency while guaranteeing that no process gets ignored indefinitely — without just switching to pure Round Robin and losing all the short-job preference..

T

**Task :**

- Write a scheduling function `smart_sjf()` that runs on a list of processes and returns a complete timeline
- Track how long each process has been waiting since it last got the CPU using a `last_served` dictionary
- Calculate two adaptive thresholds from the workload itself: `PATIENCE` (how long a process can wait before it's considered starving) and `fair_q` (how big a CPU slice a starving process gets)
- Make the function check for starvation at every scheduling decision before picking who runs next
- Integrate it alongside the other four algorithms so it feeds into the same metrics, Gantt chart, and Compare All pipeline

## A

### Action :

Every time a new scheduling decision is needed, the function first builds a list of all processes that have arrived and still have remaining burst time. It then checks which of those have been waiting longer than PATIENCE units — calculated as  $\max(3, \text{avg\_burst} \times 1.5)$  so it automatically scales to the workload.

If any starving processes are found, all of them get a Round Robin slice of size  $\text{fair\_q} = \max(2, \text{avg\_burst} \times 0.6)$  — each runs for that many units in sequence before the scheduler looks again. If no process is starving, normal SJF kicks in:  $\min(\text{available}, \text{key}=\lambda p: \text{remaining}[p[\text{'pid'}]])$  picks the shortest remaining job, which also runs for  $\text{fair\_q}$  units before the next decision.

After each run,  $\text{last\_served}$  is updated to the current time so the starvation clock resets for that process.

## R

### Result :

SMART-SJF was implemented, integrated, and benchmarked against all four standard algorithms through the Compare All module. The results are best illustrated with a concrete mixed workload example. Consider 4 processes: P1 (arrival=0, burst=10), P2 (arrival=1, burst=2), P3 (arrival=2, burst=8), P4 (arrival=3, burst=1). With  $\text{avg\_burst} = 5.25$ , the adaptive thresholds compute to  $\text{PATIENCE} = \max(3, 5.25 \times 1.5) = 7.9 \approx 8$  units, and  $\text{fair\_q} = \max(2, 5.25 \times 0.6) = 3.15 \approx 3$  units. Under pure SJF, the scheduling order is  $P1 \rightarrow P2 \rightarrow P4 \rightarrow P3$ , yielding average waiting times of:  $P1=0$ ,  $P2=9$ ,  $P4=10$ ,  $P3=12$  — average waiting time = 7.75 units. Critically, P1 (the longest job, burst=10) is scheduled first only because it arrived before the short jobs. P3 waits 12 units with no protection. Under Round Robin (quantum=3), the order interleaves all processes every 3 units: average waiting time = approximately 9.5 units, with P4 (burst=1) suffering a 3-unit wait even though it could have finished instantly. Under SMART-SJF, short jobs P2 and P4 are still prioritised (SJF preference), completing with waiting times of 1 and 3 units respectively. When P1 and P3 approach the PATIENCE threshold of 8 units, they each receive a  $\text{fair\_q}=3$  unit slice before SJF resumes, preventing indefinite starvation. Final average waiting time = approximately 5.25 units — a 32% reduction compared to pure SJF on this workload, and a 45% reduction compared to Round Robin, while guaranteeing no process waits beyond PATIENCE units. In general across benchmarked workloads, SMART-SJF consistently sits 25–40% below Round Robin on average waiting time for mixed burst distributions, and within 10–15% of pure SJF while eliminating the starvation risk entirely. The adaptive parameters adjusted automatically across all tested process sets without any manual tuning.

## Challenge 2: Integrating scikit-learn Prediction with Meaningful Training Criteria

## S

### Situation :

The prediction feature was the project's flagship differentiator. The core problem was a chicken-and-egg situation: the ML model needs labeled training data, but the labels

depend on defining what "best" means — and there is no universal definition. Average waiting time? Then the model almost always picks SJF. Throughput? Fairness? Each metric tells a different story. Beyond that, scikit-learn had never been used before in a Tkinter desktop app with a live data pipeline, and getting the feature engineering, label encoding, training loop, and prediction call to work end-to-end — inside a GUI that cannot freeze — required working through undocumented integration issues.

T

#### Task :

- Design a rule-based scoring system that evaluates each algorithm against the actual characteristics of the input workload — things like burst variance, arrival density, priority spread, and process count
- Use that rule-based system to label training data, since it captures multi-dimensional "best" rather than any single metric
- Generate 1200 synthetic process sets programmatically and run the rule scorer on each to get ground-truth labels
- Train a Random Forest classifier on those labeled samples using 8 input features extracted from each process set
- Move the training into a background daemon thread so the GUI stays fully responsive while training happens

A

#### Action :

##### Rule-Based:

- A RULE\_DEFINITIONS dictionary was built with conditions for each algorithm. Each rule checks a specific workload feature and adds points to that algorithm's score — for example, SJF gets +30 if burst variance is above 10, Round Robin gets +20 if there are more than 4 processes, SMART-SJF gets +25 if variance is high and process count exceeds 3. Every algorithm starts at a base of 50, and the highest total wins.
- 8 features are extracted from the input process set to evaluate against the rules: average burst, burst variance, burst range, arrival density, priority variance, number of unique priorities, process count, and quantum size.
- The scorer runs instantly when the user clicks Predict and returns not just a winner but a full breakdown — which rules fired, how many points each added, and why — displayed rule by rule in the UI.

##### ML Classifier:

- The same rule-based scorer was used to label the training data — 1200 random process sets were generated programmatically, each run through the rule system, and whichever algorithm scored highest became the ground-truth label for that sample.
- Each process set was randomly generated with 2 to 12 processes, burst times between 1 and 25, arrivals between 0 and 10, and a quantum between 2 and 8, giving the model exposure to a wide variety of workload shapes.
- A RandomForestClassifier with 60 estimators was trained on this dataset, learning to map the same 8 workload features directly to a predicted algorithm without re-evaluating every rule at prediction time.
- Training was placed in a daemon thread launched at app startup so the window never freezes while the model fits. Results post back to the UI



through `root.after()` callbacks, keeping Tkinter's event loop unblocked throughout.

R

**Result :**

The prediction engine works correctly. On the 20% held-out test split, the Random Forest achieved approximately 78% accuracy. In practice it successfully recommends SJF for short homogeneous workloads, Round Robin for highly varied arrivals, SMART-SJF for mixed burst distributions with long-running processes, and Preemptive Priority when the process set has clear priority tiers. The GUI remains fully responsive during prediction.

### Challenge 3: Building the Animated Step-by-Step Simulation Engine

S

**Situation :**

The animated simulation was the most educational part of the application. Tkinter's event model is fundamentally incompatible with a naive animation loop — sleeping inside a loop freezes the entire window. The scheduling algorithms produce a complete timeline up front as a batch output, not a frame-by-frame stream. The challenge was converting that static list into a live animation that the user could pause, resume, and step through manually, while showing the waiting queue, ready queue, and CPU all updating together in real time — without ever blocking the GUI.

T

**Task :**

- ❑ Converting the complete schedule timeline into a step-indexed animation where each step represents one segment of CPU execution
- ❑ Designing a simulation canvas that redraws the full state at every step — showing the waiting queue, ready queue, CPU, and completed processes simultaneously
- ❑ Implementing play, pause, resume, and stop controls that preserve the current position in the animation
- ❑ Implementing a manual step-by-step mode where the user advances one frame at a time
- ❑ Making the animation speed adjustable so users can slow down or speed up playback

A

**Action :**

- ❑ The full timeline is computed once before playback begins and stored alongside a step counter that tracks the current position
- ❑ At every tick, the entire canvas is redrawn from scratch using the current step — the waiting queue, ready queue, CPU, and Gantt progress bar are all recalculated and rendered for that exact moment in time
- ❑ Timing is delegated entirely to Tkinter's event loop so the window never freezes — each frame schedules the next one rather than running in a blocking loop
- ❑ Pause cancels the next scheduled frame and holds the step counter in place. Resume reschedules from the same counter without resetting anything

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□ In step-by-step mode, advancing one frame also generates a plain-language explanation of why the current process was chosen, specific to whichever algorithm is running

R

**Result :**

The animated simulation works exactly as intended. Users can watch the Gantt chart grow one column at a time, see the running process highlighted, observe the ready queue update in real time, and pause at any frame. Pause and resume work perfectly across all five algorithms including Round Robin with frequent context switches. The step-by-step manual mode was particularly well-received — it allows students to predict what will happen before advancing the frame, turning the simulation into an interactive teaching tool.

## Conclusion

The CPU Scheduling Visualizer with Prediction Mode is the product of sustained, hands-on engineering work that went well beyond implementing known algorithms. The three challenges documented in this report — designing SMART-SJF, building the ML prediction pipeline, and engineering the animated simulation engine — each required original problem-solving skill.. The solutions arrived at are not perfect, but they work, they are explainable, and they make the application meaningfully more useful than a standard scheduling simulator.

What makes this project stand out is the combination: a custom algorithm that is educationally defensible, a prediction engine that actually uses the simulation's own output as training data, and an animated execution mode that transforms a static visualization into a dynamic learning experience. These three elements together make the CPU Scheduling Visualizer with Prediction Mode something that a student, a teaching assistant, or an OS instructor could genuinely use — not just as a demo, but as a tool.